Chapter 12

Introduction to Silver Amalgam

"To study the phenomena of disease is to sail an uncharted sea…

While to study patients without books is not to go to sea at all…."

—Sir William Osler, 1901

Silver Amalgam

Definition: Dental amalgam (*silver amalgam* or simply *amalgam*) is a metallic, polycrystalline restorative material composed of a mixture of silver–tin alloy and mercury. Current alloys that are amalgamated with mercury are silver–tin–copper and small amounts of zinc (Zn).

Silver amalgam has been used for the restoration of teeth for well over a century.[1] It has been the most popular restorative material over that period of time, having been used in over 75% of single tooth restorations placed in the past 100 years (Fig. 12.1).

The use of silver amalgam has diminished somewhat in recent years due to

I. increased patient interest in esthetic restorative materials;
II. minimally invasive nature of the preparation steps for esthetic restorative materials; and
III. environmental concern for the potential toxicity of mercury.

Nonetheless, silver amalgam will continue to be an *important restorative material* for many years to come as

I. it will likely be especially valuable in many of the developing countries where dental caries continues to be a major problem and where resources required to deliver alternative (more sophisticated) restorative dentistry options are not available;
II. additionally, silver amalgam continues to be one of the best materials available for core build-ups of extensively damaged teeth to support crowns and fixed partial dentures; and
III. the choice of a composite resin material over amalgam may be based on the assumption that both materials will perform equally well over time for all patients. However, a growing body of evidence suggests that the risk of developing secondary caries adjacent to amalgam restorations is at least 2 times less likely than that of composite resin restorations in *high caries risk patients*, and therefore the reduction/phase-out of dental amalgam may have been premature.

History of Silver Amalgam

- Introduction to North America by the Crawcour brothers in the 1830s.
- *First amalgam war*, and the infamous 'amalgam pledge' that was finally rescinded in 1850.
- G.V. Black's publication of his formula for silver amalgam in 1896 (Fig. 12.2).
- In the 1930s, the writings of a German physician, Stock, stimulated research into the biocompatibility of silver amalgam, and specifically the effects of mercury. In what could be called the *second amalgam war*, claims and counterclaims were made, and eventually organized dentistry accepted the belief that amalgam was safe.
- The first American Dental Association (ADA) Specification (#1) for a dental material was established for silver amalgam, and Stock later went on to repudiate his own writings.
- The composition of silver amalgam did not change substantially from the publication of Black's formula until the 1960s when high-copper amalgam was introduced to the profession by Innes and Youdelis in 1963.[2]

  As subsequent clinical research would clearly demonstrate, the high-copper formulations that are almost exclusively used today are clinically superior to the conventional amalgam materials that had been used up to that time.
- Silver amalgam remains as one of the most important advances in dental materials in the past 50 years.[6]



Figure 12.1 Amalgam restorations.

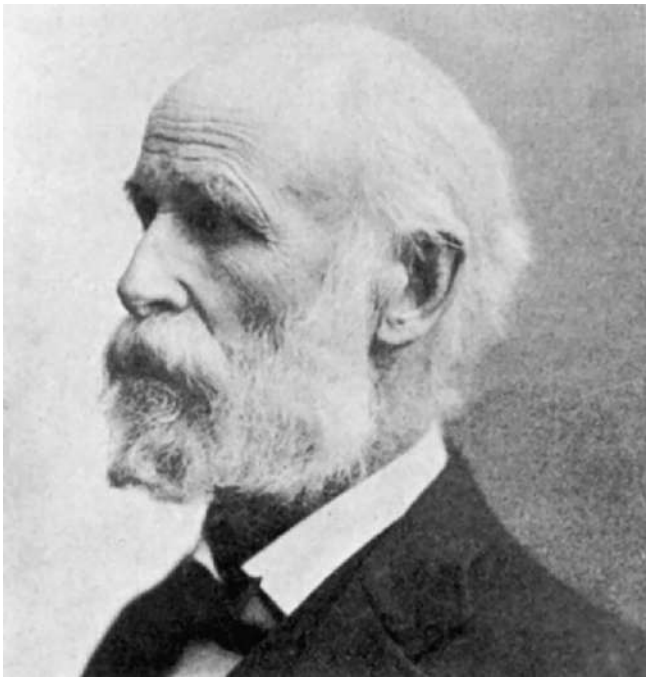


Figure 12.2 G.V. Black. Father of operative dentistry. (From Kock CRD: *History of dental surgery*, vol 1, Chicago, 1909, National Art Publishing.)

Advantages of Silver Amalgam

Silver amalgam has been and will continue to be such a popular restorative material because it possesses a number of well-defined advantages.

I. It is one of the **easiest** of all restorative materials to manipulate, it can be placed into the prepared cavity in a plastic state and then it can be carved and shaped to physiologic form before it hardens.
II. Once hardened, it has **excellent physical properties** that allow it to perform well in the long term as a restorative material in posterior teeth. It is strong enough to provide very predictable long-term service in conservative restorations, and is also valuable as an interim restorative material in very large restorations.
III. The basic strength of silver amalgam, coupled with the fact that it can be placed relatively rapidly in a single appointment, makes it a very cost-effective material. It is **less expensive** by a wide margin than any of the alternative restorative materials.
IV. Silver amalgam has one additional important advantage: its ability to automatically **seal the cavity preparation**. This ability, which is a result of the build-up of corrosion products at the interface between the amalgam and the cavity margins, makes silver amalgam unique among restorative materials. A seal can be produced with excellent technique with some of the alternative materials, but none are as user-friendly as amalgam which creates a seal independent of the operator. Several months are required before the seal becomes complete, especially with high-copper amalgam formulations; but once attained, it is a very effective barrier against recurrent caries.

Because it requires some time for the seal resulting from deposition of corrosion products to materialize, a method of obtaining a temporary seal at the time of placement is recommended. *In contemporary practice copal varnish is obsolete, and sealing the cavity with a desensitizing agent such as Gluma Desensitizer or G5 prior to amalgam placement is indicated.*

Classification of Silver Amalgam

Silver amalgam is supplied to the dentist as a powder or metallic pellets that are then mixed together with liquid mercury to form a plastic mass that can be placed into a prepared cavity. The plastic material allows sufficient working time for successful placement and then sets to form a strong and durable restoration that can provide long-term clinical service.

The major approaches to the classification of amalgams are shown in Box 12.1.

Box 12.1 Classification of amalgam

1. Based on copper content
   a. Conventional or low-copper alloy
   b. High-copper alloy
      – High-copper admixed alloy
      – High-copper unicompositional alloy
2. Based on amalgam alloy particle geometry and size
   a. Lathe-cut alloy
      – Regular-cut
      – Fine-cut
      – Microfine-cut
   b. Spherical alloy
   c. Admixed alloy
3. Based on zinc content
   a. Zinc containing alloy
   b. Zinc-free alloy
4. New amalgam alloys

I. Based on Copper Content

A. Conventional Amalgams (Low-Copper Amalgam)

- Low-copper amalgam alloys contain 2%–5% copper.
- A typical modern low-copper amalgam alloy may contain **around 70% silver, 25% tin, 4% copper and 1% zinc**.

B. High-Copper Amalgam

- High-copper amalgam alloys contain 12%–30% copper. Increasing the copper content *greater than 12% by weight* in the amalgam alloy effectively *suppresses the formation of the more corrosive gamma-2 phase (Sn–Hg)*, which is prone to intra-oral corrosion.
  - Due to their higher copper content, these alloys display significantly *better corrosion resistance* than do low-copper amalgams.
- A typical high-copper amalgam alloy may contain **70% silver, 16% tin, 13% copper and 1% zinc** (Fig. 12.3; Table 12.1).
- High-copper amalgams can be produced from amalgam alloy particles that are lathe cut or spherical.
- A dramatic improvement in corrosion resistance led to a *doubling* or *tripling* of clinical longevity of these amalgams. However, enough corrosion occurs at the amalgam–tooth interface to result in the successful *sealing* of the restoration.
- These materials can provide satisfactory performance for more than 12 years.



Figure 12.3 Dispersalloy (dispersed phase alloy) high-copper amalgam.
(Courtesy: Dentsply.)

Table 12.1 High-copper amalgam alloy composition

Composition	Percentage (%)	Characteristic
Ag	70	Strength
Sn	16	Expansion
Cu	13	Strength
Zn	1	Deoxidizer

High-copper admixed alloy

- *Innes* and *Youdelis* added silver–copper spheres to the conventional amalgam alloy, and demonstrated the advantageous effect of copper on corrosion resistance.
- This type of alloy is known as *admixed* as the final alloy is a combination of lathe-cut low-copper alloy along with spherical silver copper alloy.

High-copper unicompositional alloy

- Each particle in this type of high-copper amalgam alloy is made up of the same composition, hence the name *unicompositional* or *single composition alloy.*

Clinical Notes

- In this book, unless otherwise specified, the term *amalgam* refers to high-copper dental amalgam.

II. Based on Amalgam Alloy Particle Geometry

A. Lathe-Cut Alloy

- Traditional amalgam alloy was manufactured by grinding up the ingot of the alloy, typically on a lathe. For that reason, those particles became known as *lathe-cut particles* (Fig. 12.4A).
- The filings were *irregular in shape* and were gradually produced in progressively finer sizes by manufacturers to control the reaction, produce smoother mixtures and enhance final properties.
- Lathe-cut particles could be classified as follows:
  - Regular-cut
  - Fine-cut
  - Microfine-cut

  Lathe-cut particles pack together relatively poorly (Fig. 12.4A) and require a relatively large amount of mercury (50%–60% by weight in the mixture) to fill in the spaces.
- Mercury-rich mixtures, after trituration but before placement into the preparation, historically could be partially condensed by wringing the mass in a squeeze cloth.

B. Spherical Alloy Particles

- By spheroidizing the alloy particles, the particles could be condensed into the tooth preparation with little condensation pressure (Fig. 12.4B) and required much less mercury to make a practical mixture.
- Spherical particles also increased the fluidity of the mixture by presenting less resistance to particle sliding. Using some or all spherical alloy particles, it is possible to reduce the mercury portion of the mixture to less than 50% by weight.
- The consistency of the mix is often described as 'buttery', and the amalgam is easily placed into the cavity and adapted to the cavity walls with a minimum of pressure. However, because the fresh amalgam mix will not resist a moderate amount of condensation pressure, it is somewhat difficult to attain an adequate proximal contact in class II restorations. With experience, this deficiency can be overcome with excellent clinical technique.
  - This alloy is well suited for very large amalgam restorations such as complex amalgams or foundations. It may also allow preparation of the freshly placed amalgam 15 min after insertion in those situations where it is being used as a build-up prior to crown fabrication. The gains in final compressive strength are likely of minor importance.

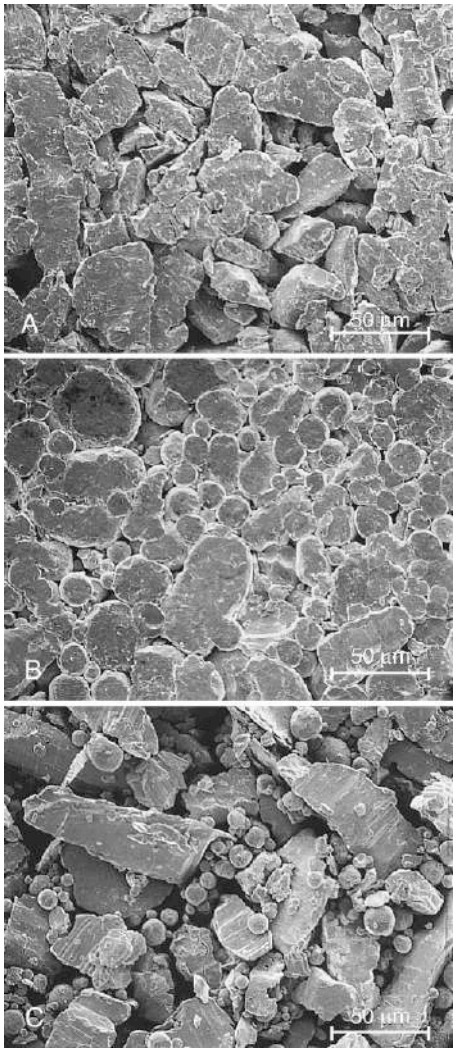


Figure 12.4 Examples of amalgam alloy powder particles. (A) Filings (New True Dentalloy). (B) Spheres (Cupralloy). (C) Mixed geometries (Dispersalloy).
(Courtesy of S.C. Bayne, School of Dentistry, University of Michigan, Ann Arbor, MI.)

C. Admixed Alloy Particles

- With admixed alloys, small spheres formed by atomizing the liquid alloy are added to the mix in the hope that a denser final amalgam will result. Such an alloy is termed an admixed alloy, and is the *most common amalgam alloy material in use today*.
- The concept is to have an alloy, which would combine the advantages offered by both the lathe-cut and spherical alloys (Fig. 12.4C).
- The irregular shape of many of the particles makes a mass that requires more condensation pressure (which many dentists prefer) and permits this heavier condensation pressure to assist in displacing matrix bands to generate proximal contacts more easily.
  - Strength of these alloys is time-dependent, and 24-h compressive strength is considerably greater than 1-h strength (Table 12.3). These alloys should be carved and burnished, and final polishing should be delayed for at least 24 h.

III. Based on Zinc Content

Concept

Zinc was added to conventional amalgams as a processing aid to *suppress oxidation* of the key elements in the alloy. Zinc tends to oxidize preferentially, forming a zinc oxide film that covers the surface of liquid alloy during manufacture and suppresses oxidation of other elements.

A. Zinc-Containing Alloy

They contain 1% *or more of zinc* in the final composition. A detrimental side effect of this residual zinc was that moisture contamination before setting converted the zinc to zinc oxide and produced *hydrogen gas* that could expand the amalgam excessively, causing pain to the patient. This type of expansion is known as **delayed expansion** or *secondary expansion*.

B. Zinc-Free Alloy

Manufacturers also produced non-zinc amalgams as an alternative. These alloys often were favoured in cases in which isolation was difficult. *Proper isolation* to prevent moisture contamination during amalgam manipulation and condensation of the zinc-containing alloy prevents the problem of delayed expansion.

Clinical Notes

- Clinical research evidence supports that zinc-containing, low-copper and high-copper amalgams may last *20%–50% longer* than do zinc-free amalgams.

IV. New Amalgam Alloys

Due to the concern about mercury toxicity, many new compositions of amalgam are being promoted as *mercury-free* or *low-mercury* amalgam restorative materials.

Alloys with **gallium** or **indium** or alloys using cold-welding techniques are presented as alternatives to mercury-containing amalgams.

None of these new alloys show sufficient promise to become a universal replacement for current amalgam materials.

Clinical Recommendations for Choosing Amalgam Alloy

I. Many dentists prefer an *admixed high-copper zinc-containing alloy* for the majority of conventional restorations where a dry field can be maintained and where achieving good interproximal contact is desirable.

II. They would choose a *spherical high-copper zinc-containing alloy* in those situations where

  i. This type of alloy is also desirable when building up an extensively broken down tooth prior to preparing that tooth for a cast crown. Here the high early strength is useful, as is the ease of condensation especially when small slots, potholes and small undercut areas are utilized as the primary retentive features; and

  ii. A dry field can be maintained but using strong condensation force is difficult, for example when restoring a class V lesion on the distobuccal of a maxillary second molar.

Setting Reaction of Amalgam

The principal considerations for any amalgam are the amount of mercury in the final restoration and the types of reaction products formed. Schematically, the basic reaction is summarized in a simple way in Fig. 12.5 and Box 12.2. The various phases involved during the setting of low-copper and high-copper alloys are tabulated in Table 12.2.

Clinical Notes

Since the residual alloy particles have physical, chemical and mechanical properties that are significantly better than those of the reaction products, it is important to *minimize the amount of matrix* that forms during the reaction.

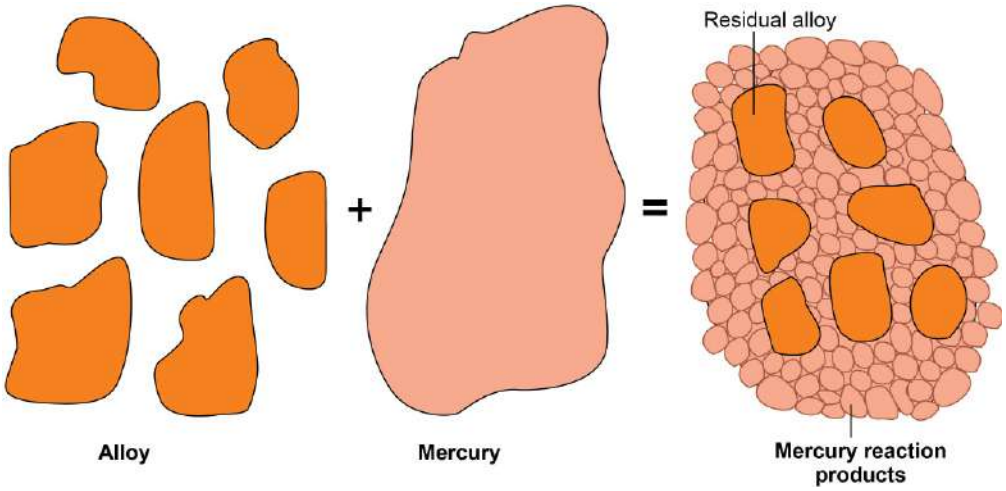


Figure 12.5 Schematic summary of setting reaction of amalgam and its associated microstructure. Before reaction, alloy particles are dispersed in mercury. After reaction, residual alloy particles are embedded in a matrix of crystalline reaction products. Only a small percentage of individual powder particles is required to react completely with mercury.

Box 12.2 Basic setting reaction of amalgam

Alloy particles of amalgam + Mercury → Dental amalgam + Unreacted alloy particles

Since the original mixture contains a large excess of silver–tin alloy particles, only a minor portion of the outside of the particles is consumed during the reaction with mercury.

↓

The unreacted portion of the original amalgam alloy particles remains as residual alloy particles, reinforcing the final structure.

↓

The reaction products form a matrix surrounding the residual alloy particles.

Table 12.2 Phases of amalgam

Silver–tin phase	γ (Gamma)	Ag_3Sn
Silver–mercury phase	γ_1	Ag_2Hg_3
Tin–mercury phase	γ_2	$Sn_{7-8}Hg$
Copper–tin phase	ε (Epsilon)	Cu_3Sn
Copper–tin phase	η (Eta)	Cu_6Sn_5

Conventional Low-Copper Amalgam

This type of amalgam sets by the reaction of silver–tin from silver–tin particles with mercury to produce two reaction product phases: (i) the silver–tin phase *gamma-1 phase* (γ_1) and (ii) the tin–mercury phase *gamma-2 phase* (γ_2) (Box 12.3).

Gamma Phase (γ)

- The major reaction product phases are silver–mercury and tin–mercury (approximately Ag_2Hg_3 and $Sn_{7-8}Hg$).
- The original alloy is designated as *gamma phase* (γ), and the reaction product phases are called *gamma-1* (γ_1) and *gamma-2* (γ_2).

Gamma-1 Phase (γ_1)

The silver–mercury (γ_1) crystals are generally small and forms most of the matrix. This phase has intermediate corrosion resistance.

Gamma-2 Phase (γ_2)

- Tin–mercury (γ_2) reaction product crystals are *long and blade-like*, penetrating throughout the matrix. Although they constitute less than 10% of the final composition, they form a penetrating matrix because of inter-crystalline contacts between the blades (Fig. 12.6).

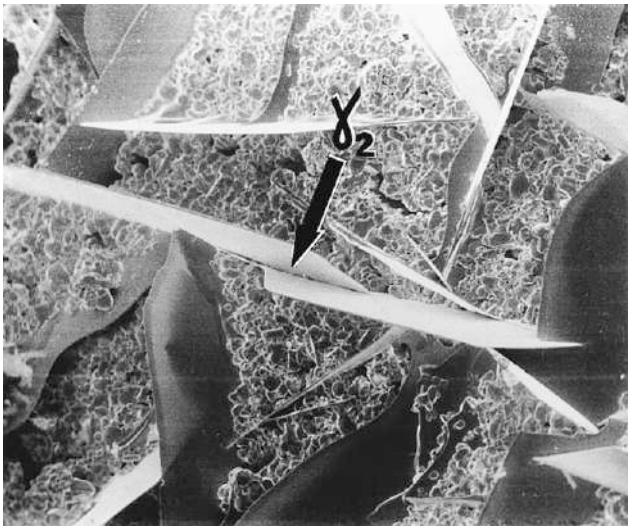


Figure 12.6 Scanning electron microscopic view of tin–mercury (Sn–Hg) (γ_2) crystals that occur in a matrix of set low-copper amalgams. Note the blade-like crystals that penetrate amalgam and touch each other to create a continuous matrix *(arrow)*.
(Courtesy of D.F. Taylor, School of Dentistry, University of North Carolina, Chapel Hill, NC.)

- This phase is *prone to corrosion* in clinical restorations, a process that proceeds from the outside of the amalgam, along the crystals, connecting to new crystals at inter-crystalline contacts.
- This process produces penetrating corrosion that generates a porous and spongy amalgam with minimal mechanical resistance. Two key features of this degradation process are as follows:
  - The corrosion-prone character of the tin–mercury phase *gamma-2 phase* (γ_2).
  - The connecting path formed by the blade-like geometry of the crystals. Both these are eliminated by the use of more copper in the initial composition.

Clinical Notes

The tin–mercury gamma-2 phase (γ_2) is the weakest phase in dental amalgam and is responsible for the corrosion process.

High-Copper Amalgam

With amalgam formulations, where the alloy contains 9%–30% copper, the copper will preferentially react with the tin and thus prevent formation of the gamma-2 phase. High-copper amalgams set in a manner similar to low-copper amalgams except that tin–mercury reactions are suppressed by the preferential formation of copper–tin phases instead.

Box 12.3 Setting reaction of conventional low-copper amalgam alloy

$$\underset{\gamma}{Ag_3Sn} + Hg \rightarrow \underset{\gamma_1}{Ag_2Hg_3} + \underset{\gamma_2}{Sn_8Hg} + \text{unreacted } Ag_3Sn$$

Again, the elimination of this gamma-2 phase strengthens the resultant amalgam and is clinically manifest in the following:

- They display *less creep* than conventional low-copper amalgams.
- The main clinical manifestation of this superiority is *improved edge strength* which results in a lack of marginal ditching over the long term.
- The gamma phase, which consists of unreacted silver and tin, is the major component of the unreacted alloy and is also present in the set amalgam. It is suspended in a matrix of gamma-1, which is the reaction product of silver and mercury. The gamma phase is very strong and corrosion resistant, whereas the gamma-1 phase is less strong and is more corrosion susceptible. The gamma-2 phase, the reaction product of tin-mercury, is also suspended in the gamma-1 matrix. It is the weakest and most corrosion susceptible of the three phases. The major difference between conventional and high-copper amalgams is the near elimination of the gamma-2 phase in the high-copper amalgams. Elimination of this phase produces a much stronger and clinically superior restoration.

High-Copper Admixed Alloy

In high-copper admixed alloys, the reaction takes place in two steps (Box 12.4). There is elimination of gamma-2 phase, which is the weakest phase.

High-Copper Unicompositional Alloys

The setting reaction of unicompositional alloy also eliminates the formation of the weak gamma-2 phase (γ_2) and the phases involved are shown in Box 12.5. Copper–tin phases that are part of the set amalgam matrix are *much less corrosion-prone* than are the tin–mercury phases they replace.

Box 12.5 Setting reaction of unicompositional high-copper amalgam alloy

$$\underset{}{Ag_3Cu_3Sn} + Hg \rightarrow \underset{\gamma_1}{Ag_2Hg_3} + \underset{\eta}{Cu_6Sn_5} + \text{unreacted } Ag_3Cu_3Sn$$

Clinical Notes

- The incorporation of higher percentage of copper in the alloy in the admixed and unicompositional alloy *eliminates* the formation of the weakest gamma-2 phase (γ_2).

Dispensing of Amalgam Alloy and Mercury

The quality of a given mix of silver amalgam is controlled by the quality of the product as supplied to the dentist, and can be modified by the manipulation of that product by the dentist. The manipulative variables include the following.

Mercury/Alloy Ratio

In general terms, the less mercury that remains in the final restoration, the better it will be both in terms of strength and corrosion resistance. Thus, the initial mix should be made with only as much mercury as is required to make an adequate mix. In the 1960s, Eames was the first to promote a low mercury-to-alloy mixing ratio (***Eames technique*** or *no-squeeze-cloth technique*). *The mercury:alloy ratio in this technique is 1:1* and is recommended for lathe-cut alloys.

I. With most admixed alloys, this will be about 50% mercury.
II. With spherical alloys, it will be slightly lesser.

Box 12.4 Setting reaction of high-copper admixed alloy

Step 1

$$\underset{\gamma}{Ag_3Sn} + Ag-Cu + Hg \rightarrow \underset{\gamma_1}{Ag_2Hg_3} + \underset{\gamma_2}{Sn_8Hg} + Ag_3Sn + Ag-Cu$$

Step 2

$$\underset{\gamma_2}{Sn_8Hg} + Ag-Cu \rightarrow \underset{\gamma_1}{Ag_2Hg_3} + \underset{\eta}{Cu_6Sn_5} + \text{unconsumed } Ag-Cu \text{ and } Ag_2Hg_3$$

Trituration

Mixing the amalgam is called trituration. Adequate trituration of the amalgam is essential to produce a workable mass and to ensure optimum physical properties. Modern amalgams are produced from pre-capsulated alloy and mercury. The components are separated in the capsule by a special diaphragm that is broken when the capsule is 'activated' just before mixing.

Pre-capsulated (pre-proportioned) amalgam (Fig. 12.7) provides convenience and some degree of assurance that the materials will not be contaminated before use or spilled before mixing.

Amalgamator

- During the early part of the 20th century, alloy powder and mercury were proportioned crudely and mixed manually. To proportion and mix amalgam more carefully, manufacturers later recommended the use of amalgamators.
- The typical amalgamator has been designed to grasp the ends of the capsule in a claw that is oscillated in a *figure-of-eight pattern*. This design accelerates the mixture towards each end of the capsule during each throw and impacts the mixture with the pestle.
  - Each brand of amalgam has specific instructions regarding optimum time and speed of trituration. The instructions are specific for each brand of amalgamator.

  To guarantee that the amalgam alloy and mercury are mixed efficiently and consistently, it is important to *calibrate amalgamators* periodically.
- Electronic amalgamators (Fig. 12.8) have digital controls and permit programming of *trituration speed* and *trituration time* settings.

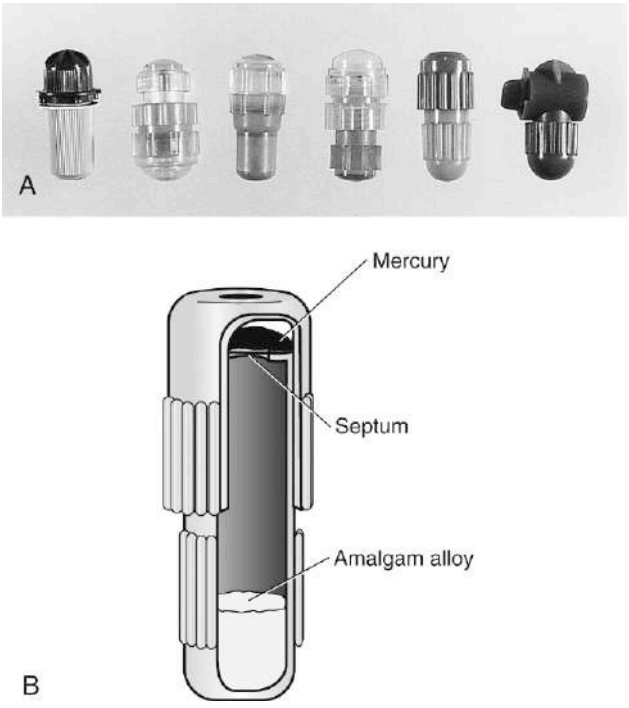


Figure 12.7 Pre-proportioned alloy and mercury in pre-packaged capsules (pre-capsulated) for mixing amalgam constituents using an amalgamator. (A) Examples of pre-proportioned capsule designs. (B) Schematic diagram of pre-proportioned capsule showing mercury and powder separated by the septum that must be perforated before mixing.
(Part A: From Rinne VW: Aluminum foil pouch packaging in premeasured amalgam capsules, *J Dent Res* 62:116–117, 1983; part B: from Daniel SJ, Harfst SA, Wilder RS: *Mosby's dental hygiene: concepts, cases, and competencies*, ed 2, St. Louis, MO, 2008, Mosby.)



Figure 12.8 Examples of dental amalgamator for automatically mixing amalgam in capsules.
(Courtesy of Dentsply International, York, PA.)

Clinical Notes

The longer and the faster an amalgam is mixed, the more rapidly it will set. The manufacturers' guidelines for mixing should be followed and can be modified slightly to obtain a faster or slower setting material.

Condensation

Condensation of the amalgam is probably the *most critical manipulative variable*. Adequate condensation results in intimate adaptation of the amalgam to the cavity walls, as well as a denser material to maximize strength. The prepared cavity is always overfilled and carved back to proper anatomic contour. This is because condensation results in a mercury-rich superficial layer, which is removed during the carving procedure. The resultant amalgam surface has considerably less mercury and subsequently superior physical properties.

The most common error made by dentists is *under condensation*. This results in poor adaptation to the cavity walls, increased micro-leakage and recurrent caries. The final amalgam is weaker and more fracture prone, and cannot be polished nearly as well as a properly condensed amalgam.

Carving and Finishing of the Restoration

With the high-copper formulations currently in use, both *pre-carve* and *post-carve burnishing* is recommended. This is generally accomplished with a large, egg-shaped burnisher. The amalgam is carved after achieving a firm set, with careful attention being paid to eliminating all marginal flash and carving the amalgam flush with the margins of the cavity preparation. Any marginal flash will eventually fracture, and it is rare that it fractures right at the margin. Inevitably, marginal ditching will occur.

After carving back to anatomic form, the restoration is lightly burnished to a smooth sheen. The *restoration may be polished after 24 h*. There are no solid clinical data that support the routine polishing of amalgam restorations, but certainly highly polished restorations are more comfortable for the patient and will be less likely to retain bacterial plaque. Polishing amalgams also indicates a certain amount of pride in workmanship and is a source of great personal satisfaction.

Bonded Amalgams

Bonded amalgams utilize any one of the current generation of enamel/dentin bonding agents to seal the dentin and 'bond' the amalgam to the prepared tooth. The amalgam is condensed into polymerizing dual cure resin that is placed on conditioned dentin and etched enamel. The amalgam or the oxide of amalgam will chemically bond to the resin, and the resin is bonded to the tooth structure primarily by micro-mechanical retention. Thus, the resin will partially retain the amalgam in place, reducing the need to create retentive undercuts in the tooth. Elimination of these undercuts, which have been shown to weaken cusps, coupled with the potential strengthening effect may improve the long-term prognosis of these restorations. Clinical research thus far has shown that bonded and non-bonded amalgams perform equally well. There seems to be little advantage to bonding amalgams, and the technique is more complex and time-consuming.[3]

This textbook does not recommend the clinical usage of bonded amalgam restorations.

Clinical Notes

I. Silver amalgam has been an important restorative material for many years and will continue to be an important material for years to come.
II. Advantages of silver amalgam are excellent physical properties, cost, long-term service and the ability to self-seal the cavity.
III. Amalgam is the most user-friendly material in restorative dentistry.
IV. High-copper amalgams are superior to conventional amalgams demonstrating excellent marginal integrity over time. This is due to elimination of the gamma-2 phase (tin–mercury).
V. It is easier to create a good proximal contact with admixed versus spherical alloys.
VI. Spherical alloys can be prepared 15 min after insertion and are indicated for amalgam core build-ups.
VII. Proper condensation of the amalgam is the most important manipulative variable.
VIII. Bonding amalgam to tooth structure is possible, but there is minimal clinical benefit.

Clinical Considerations

Strength

Compressive Strength

- The compressive strengths of high-copper amalgams are greater than the strengths of low-copper amalgams because of the presence of the copper phases.
- High-copper amalgams have compressive strengths that range from **380 to 550 MPa** (55,000–75,000 psi) and are similar to those of enamel and dentin (Table 12.3).

Tarnish and Corrosion

Low-copper and high-copper amalgams undergo two kinds of corrosion:

I. Chemical corrosion (tarnish)
II. Electrochemical corrosion

Table 12.3 Compressive strength (MPa) of different brands of amalgam alloys after 1 and 24 h

Alloy	1 h	24 h
Velvalloy	18,000	55,000
Dispersalloy	33,000	64,500
Phasealloy	12,900	65,700
Tytin	34,900	75,000
Valiant	47,800	76,500

Chemical corrosion (tarnish)

- This occurs most notably on the occlusal surface and produces a *black film of silver–sulphur (Ag–S) tarnish*.
- This discoloration is *limited to the surface* and does not compromise any properties except for esthetics.

Electrochemical corrosion

The basic components required for electrochemical corrosion to take place are (i) an anode (site of corrosion), (ii) a cathode, (iii) a circuit and (iv) an electrolyte.

This kind of corrosion is due to the oxidation of the tin–mercury phase of amalgam leading to the formation of oxides and oxychlorides of tin:

$$\underset{\text{(Oxidation)}}{\text{Tin – mercury phase}} \rightarrow \underset{\text{(Tin – oxygen)}}{\text{Sn – O}} + / \text{ or } \underset{\text{(Tin – oxygen – chlorine)}}{\text{Sn – O – Cl}}$$

The oxide precipitates as crystals and tends to fill up the spaces occupied by the original tin–mercury phase. Along the margins of the amalgam, tin–oxygen helps seal the space against micro-leakage (Fig. 12.9).

The mechanism of electrochemical corrosion is explained in Box 12.6, and this kind of amalgam corrosion has the potential to occur virtually anywhere on or within a set amalgam. The following are the various kinds of electrochemical corrosion (Fig. 12.10).

Galvanic Corrosion

If an amalgam is in direct contact with an adjacent metallic restoration such as a gold crown, the amalgam is the anode in the circuit. This type of electrochemical corrosion is called *galvanic corrosion* and is associated with the presence of *macroscopically* different electrode sites.

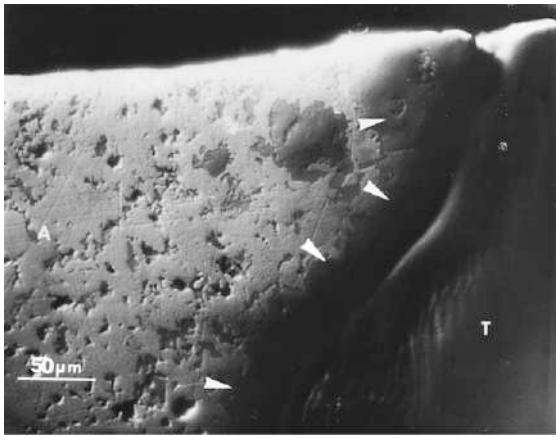


Figure 12.9 Marginal sealing by corrosion products. Scanning electron microscopic cross-sectional view of tin–oxide (Sn–O) corrosion products sealing the amalgam (*A*) margin along the enamel wall (*T*).
(From Port RM, Marshall GW: Characteristics of amalgam restorations with variable clinical appearance, *J Am Dent Assoc* 110:491–495, 1985.)

Box 12.6 Mechanism of electrochemical corrosion

Electrochemical corrosion occurs whenever *chemically different sites* act as anode and cathode.

↓

This corrosion requires the sites be connected by an electrical circuit in the presence of an electrolyte, typically *saliva*.

↓

The anode corrodes, producing soluble and insoluble corrosion reaction products.

Local Galvanic Corrosion

The same process may occur *microscopically* (*local galvanic corrosion*) because of the electrochemical differences of different phases within the amalgam. Residual amalgam alloy particles act as the strongest cathodes. Tin–mercury or copper–tin reaction product phases are the strongest anodes in low-copper and high-copper amalgams.

Concentration Cell Corrosion or Crevice Corrosion

Local electrochemical cells may also arise whenever a portion of the amalgam is covered by *plaque* or soft tissue. The covered area has a locally lowered oxygen or higher hydrogen ion concentration, making it behave more anodically and corrode. Cracks and crevices produce similar conditions and preferentially corrode.

Stress Corrosion

Regions within an amalgam that are under stress also display a greater propensity for corrosion (stress corrosion).

Clinical significance

Amalgam has a linear coefficient of thermal expansion that is 2.5 times greater than the tooth structure, and it does not bond to the tooth structure. During expansion and contraction, percolation occurs along the external walls (Fig. 12.11). The formation of corrosion products prevents the fluid ingress and egress along the margins and contributes to the *self-sealing ability of amalgam*.

Clinical Notes

- Electrochemical corrosion is not a mechanism of mercury liberation from set amalgam.
- Mercury immediately reacts with locally available silver and tin from residual amalgam alloy particles and is re-consumed to form more corrosion reaction products.

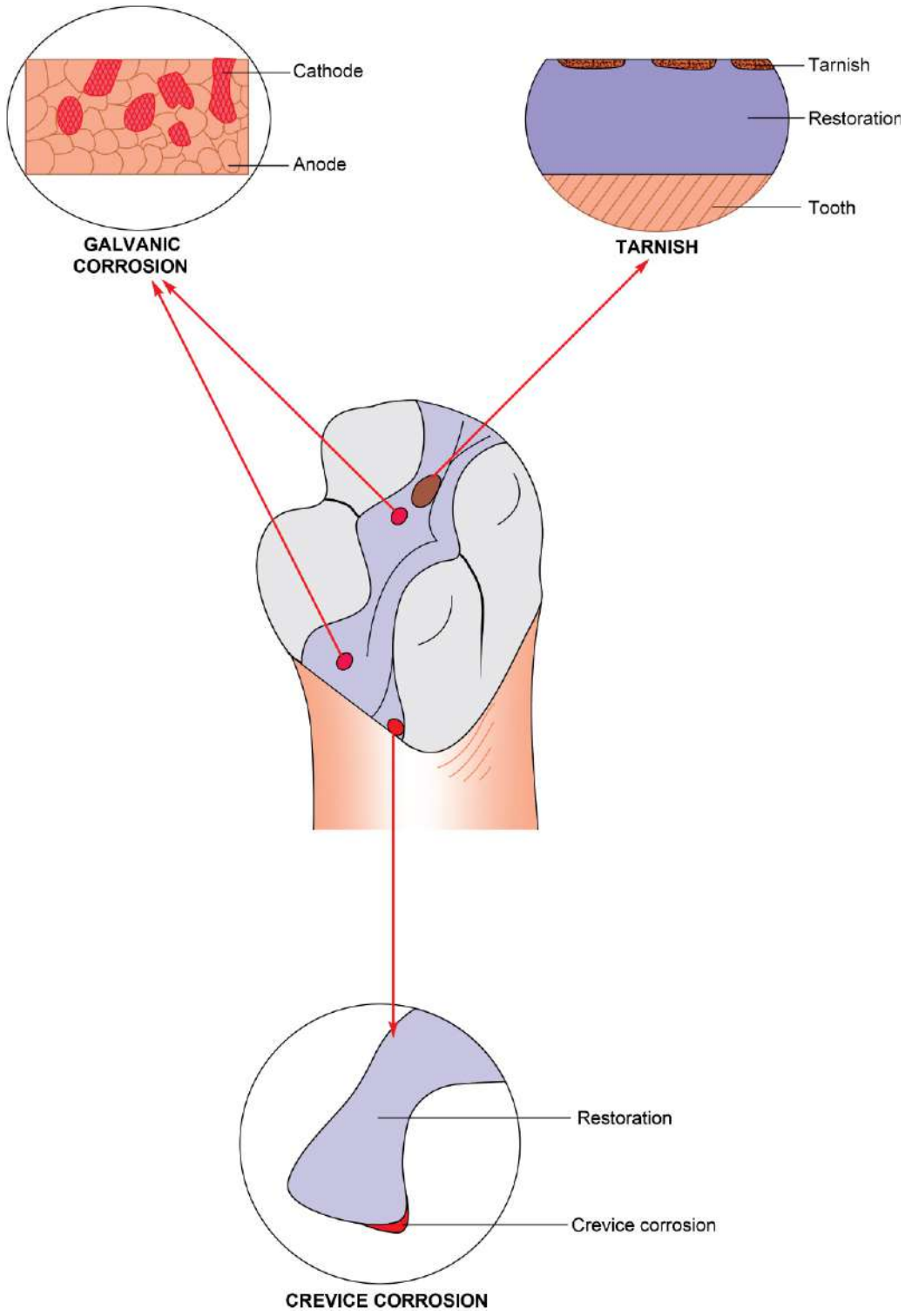


Figure 12.10 Examples of sites susceptible to electrochemical and chemical corrosion on amalgams: galvanic corrosion at interproximal contact with metallic restoration, such as gold casting alloy; local galvanic corrosion on occlusal surface at grain boundaries between different metallic phases; crevice corrosion at margins owing to lower pH and oxygen concentration of saliva and under retained interproximal plaque owing to lower local pH; chemical corrosion of occlusal surface with sulphide ions in saliva, producing surface tarnish.

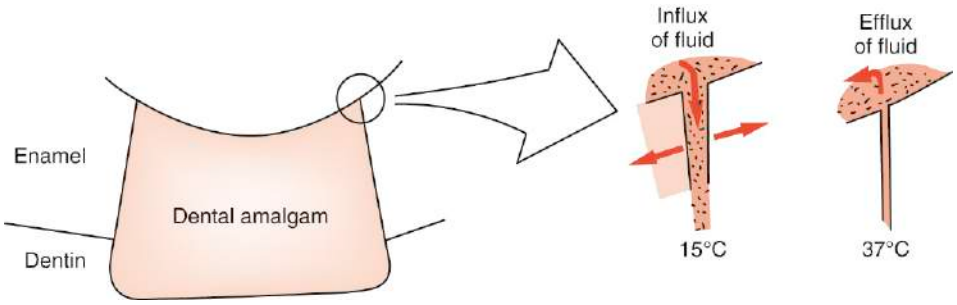


Figure 12.11 Percolation along the margins of an amalgam restoration owing to its difference in linear coefficient of thermal expansion from tooth structure during intra-oral temperature changes. Fluid influx occurs during cooling (contraction). Fluid efflux occurs during heating (expansion).

Creep

Definition: Amalgam creep is defined as the time-dependent plastic deformation of zinc-containing amalgam alloys principally resulting from slow metallurgic phase transformations that involve diffusion-controlled reactions and produces volume increase.

Clinical significance

I. The associated expansion makes amalgam *protrude* from the tooth preparation. Such secondary expansion can occur throughout the clinical life of a restoration.
   i. **On non-occlusal surfaces**: The entire amalgam restoration may appear *extruded* (Fig. 12.12), and this can produce esthetics-related problems or overhangs in some areas.
   ii. **On occlusal surfaces**: Abrasion and attrition tend to limit the overall extrusion. Occlusal margins become *fracture-susceptible* ledges elevated above the natural contours of the adjacent enamel (Fig. 12.13).

II. Extrusion at margins is promoted by *electrochemical corrosion*, during which mercury from tin–mercury reacts with silver–tin particles and produces further expansion during the new reaction. This mechanism, called *mercuroscopic expansion*, originally was proposed by Jorgensen as an explanation for the prevalence of marginal fracture associated with occlusal amalgams.

III. The most common evidence of degradation of low-copper amalgams is *marginal fracture* (Fig. 12.14).

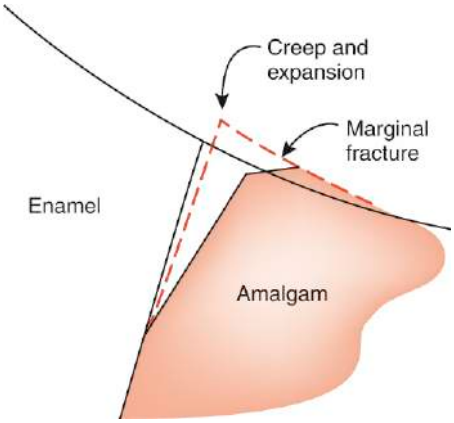


Figure 12.13 Schematic view of class I amalgam restoration that was extruded by mercuroscopic expansion, underwent marginal fracture and now contains marginal ditch.
(Courtesy of S.C. Bayne, School of Dentistry, University of Michigan, Ann Arbor, MI.)



Figure 12.12 Clinical photograph of class V amalgam restoration being extruded by mercuroscopic expansion.



Figure 12.14 Occlusal amalgam restoration with extensive marginal deterioration.
(From V. Gopikrishna: *Preclinical manual of conservative dentistry*, ed 2, 2014, Elsevier.)

Mercury and Silver Amalgam

Silver amalgam contains between 43% and 52% mercury when it is freshly mixed, prior to condensation into the prepared cavity. The procedures of condensation, pre-carve burnishing and carving removes a portion of this mercury. The remainder of the mercury remains in the finished restoration. Because high doses of mercury vapour can cause serious systemic illness, this mercury remaining in silver amalgam restorations has been a source of continued controversy for over a century.[1,4]

In the 1990s, renewed controversy surrounding amalgam resulted in what might be called the *third amalgam war*. This particular episode in the turbulent history of amalgam has been precipitated by a combination of renewed scientific investigation coupled with an unprecedented amount of media hype.[5]

This controversy continues unabated today despite a total lack of scientific evidence demonstrating a link between silver amalgam and systemic disease. In fact, quite to the contrary, ample scientific data support the continued use of silver amalgam and document its enviable record of safety.

When the amalgam setting reaction is complete, only extremely minute levels of mercury can be released, and those are far below the current health standard. The important perspective is that mercury enters the body everyday, no matter what restorative filling materials are present in the mouth, and is biochemically processed and *excreted safely*. Careful review of amalgam-handling procedures reveals that the critical times are when metallic mercury exists in *liquid* or *vapour form*, rather than bound in a set amalgam.

Clinical Recommendations

Specific dental mercury hygiene recommendations are summarized in Box 12.7.[6–8] Fig. 12.15 summarizes all of the potential mercury management problems.

In the dental office, the sources of mercury exposure related to amalgam and the best practice guidelines include the following:

I. *Amalgam raw materials being stored for use (usually as precapsulated packages)*:
   i. Mercury-containing products should not be stored in the open, but rather in closets or cabinets, to minimize local concentrations in the rest of the office.
   ii. Storage locations should be near an exhaust vent that carries air out of the building.

II. *Mixed but unhardened amalgam during trituration, insertion and intra-oral hardening*:
   i. Small local spills or spatters of triturated materials are best dealt with by collection with a vacuum aspirator (not a vacuum cleaner).
   ii. To minimize the risk of mercury aerosol or vapour during trituration, small covers are mounted on mechanical triturators to contain the aerosol to the

Box 12.7 Dental mercury hygiene recommendations

(From the American Dental Association Council on Scientific Affairs: Dental mercury hygiene recommendations, *J Am Dent Assoc* 130:1125–1126, 1999.)

1. *Educate* all personnel involved in the handling of mercury or dental amalgam on the potential hazard of mercury vapour and the necessity for observing good mercury hygiene practices.
2. Make personnel aware of the *potential sources of mercury* vapour in the dental operatory (e.g. spills; open storage of amalgam scrap; open storage of used capsules; trituration of amalgam; placement, polishing or removal of amalgam; heating of amalgam-contaminated instruments; leaky capsules or bulk mercury dispensers).
3. Work in *well-ventilated spaces* with fresh air exchanges and outside exhaust. If the spaces are air-conditioned, air-conditioning filters should be replaced periodically.
4. *Regularly check* the dental operatory atmosphere for mercury vapour. *Dosimeters* and *mercury* vapour *analyzers* may be used for monitoring. The current limit for mercury vapour established by the Occupational Safety and Health Administration (OSHA) is *50 μg/m³ (time-weighted average) in any 8-h work shift over a 40-h work week.*
5. *Floor coverings* should be non-absorbent, seamless and easy to clean.
6. Use *only pre-capsulated alloys*; discontinue the use of bulk mercury and bulk alloy.
7. Use an *amalgamator* with a completely enclosed arm.
8. Use care in handling amalgam. *Avoid skin contact* with mercury or freshly mixed amalgam.
9. Properly *dispose* of single-use capsules according to applicable waste disposal laws.
10. Use *high-volume evacuation when finishing or removing amalgam*. Evacuation systems should have traps or filters. Check and clean or replace traps and filters periodically to remove waste amalgam (including contact amalgam) from the waste stream.
11. Salvage and store all scrap amalgam (i.e. noncontact amalgam remaining after a procedure) in a tightly closed container, *either dry or under radiographic fixer solution*. Amalgam scrap should not be stored in water. If the scrap is stored dry, mercury vapour can escape into room air when the container is opened. If the scrap is stored under radiographic fixer solution, special disposal of the fixer may be necessary. Some recyclers only accept scrap amalgam that is dry. When feasible, recycle amalgam scrap and waste amalgam. Otherwise, dispose of amalgam scrap and waste amalgam in accordance with applicable laws.
12. Dispose of mercury-contaminated items in sealed bags according to applicable regulations.
13. Do not dispose of mercury-contaminated items in regulated (medical) waste containers or bags or along with waste that will be incinerated.
14. Clean up spilled mercury properly using trap bottles, tape or freshly mixed amalgam to pick up droplets, and commercial cleanup kits. *Do not use a household vacuum cleaner.*
15. Remove *professional clothing* before leaving the workplace.

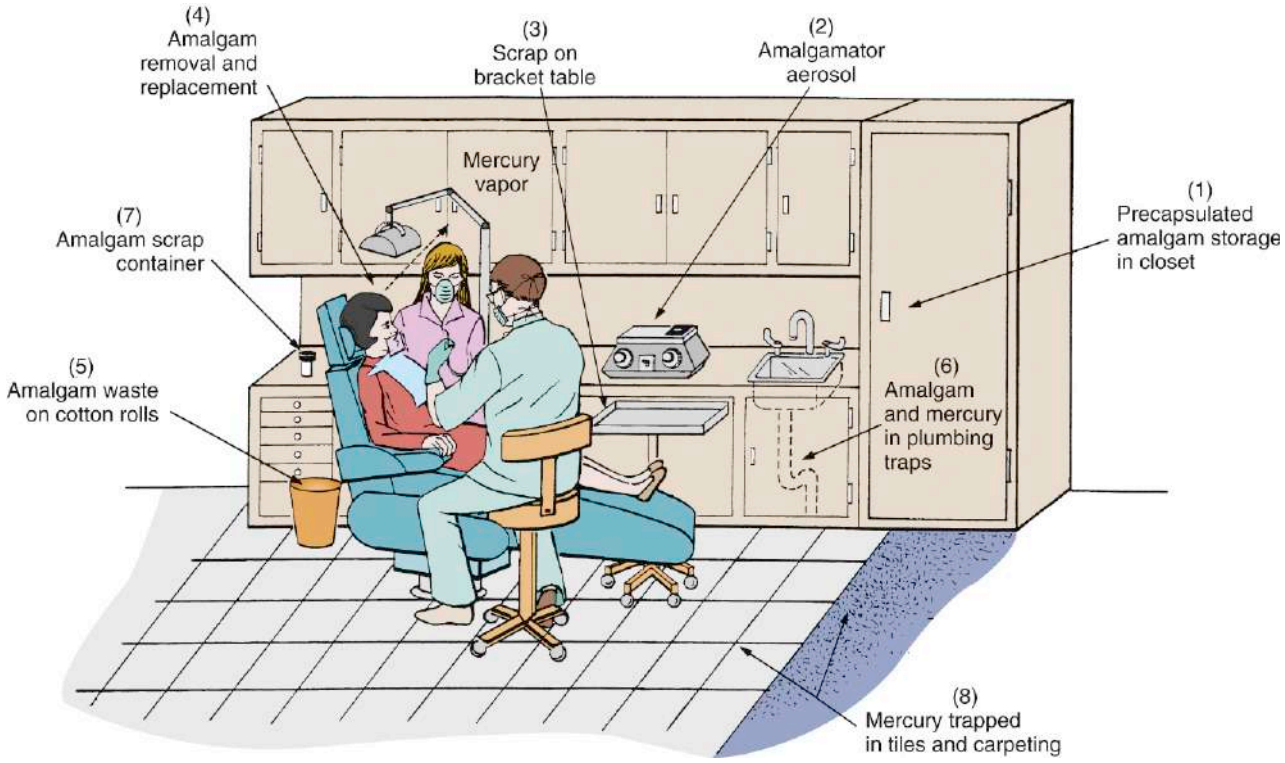


Figure 12.15 Sources of mercury hazards in the dental operatory: (1) some mercury vapour released from stored materials; (2) small losses from capsules during trituration; (3) spillage during manipulation for tooth restorations; (4) some vapour exposures to the dentist, the assistant and the patient during removal, placement or finishing or polishing of amalgam; (5) contamination of cotton rolls; (6) collection of debris via vacuum suction into the plumbing system and the sewer system; (7) collection of remnants in a jar for recycling; and (8) mercury trapped in small cracks between floor tiles or in carpet fibres.

region of the triturator. However, this does not eliminate the hazard.

iii. Air contamination is managed by ensuring that air flow is reasonably high and that fresh air is brought into the office.

III. *Amalgam scrap that has insufficient alloy to consume the mercury present completely*:

i. Scrap amalgam from condensation procedures should be collected and stored under water, glycerin or spent X-ray fixer in a tightly capped jar.

ii. When small droplets of mercury-rich material contaminate the floor coverings, the only practical approach to decontaminating the area is to replace the coverings.

iii. No effective treatment exists for removing liquid mercury from carpeting.

IV. *Amalgam undergoing finishing and polishing operations*:

i. Dentists should slowly polish amalgams using adequate cooling water in order to minimize the liquefaction of silver-mercury reaction product, which has a low melting point (127°C).

V. *Amalgam restorations being removed*:

i. Rubber dam, high-volume evacuation and water cooling are recommended during the removal of amalgam restorations as the high temperatures generated during the removal process causes the melting of the silver–mercury phase.[9]

ii. There is absolutely no scientific evidence to indicate that silver amalgam restorations pose a health risk of any kind to patients. There is a large body of well-conducted and controlled scientific literature to support this statement.

This statement may be made with absolute assurance when comparing the exposure of mercury that would occur from silver amalgam to that allowed a worker who has occupational exposure to mercury vapour. As with other potentially toxic materials, a safe level of mercury vapour that can be tolerated without untoward effects has been established by industrial safety boards.

That level is known as the *threshold limit value* (TLV). This level has been adjusted downwards over the years as more and more is known about mercury vapour and its effects. That level is currently either **50 μg/m^3** (U.S. Occupational Safety and Health Administration [OSHA]) or **25 μg/m^3** (World Health Organization [WHO]) and represents the amount of mercury vapour that can be safely allowed in the air for workers who spend 8 h/day, 40 h/week in that environment. *The worst dose a patient could receive from 10 or 12 amalgam restorations would be about 1/100 of the current TLV.*[10]

The dose of mercury vapour to a patient from 10 or 12 amalgam restorations is extremely low, and has virtually no clinical consequences. To put it into perspective, the amount

of mercury vapour a patient could absorb per day from his or her amalgams would be 1/10–1/20 of that which they receive from a normal healthy diet.[11] Recently, calculations related to dose have estimated that for the most sensitive of patients to demonstrate even minor symptoms of mercury poisoning, patients would have to have 450–500 amalgam restorations in place.[12] Of course, this is patently impossible and illustrates that the doses of mercury vapour received by patients from their amalgam restorations are indeed miniscule.

Replacement of Existing Amalgam Restorations

The widespread practice of removing functional amalgam restorations in the name of patient safety is to be deplored. This results in unwarranted loss of tooth structure, unnecessary expense and, especially if the replacement restoration is tooth coloured, limited longevity.

Unwarranted removal of amalgam restorations in order to prevent or cure systemic health conditions is wrong and in fact is considered unethical in many jurisdictions.

If required amalgam restorations can be safely removed without any hazard with the use of a *rubber dam* and *high-volume evacuation* as the patient, the dentist, the assistant and the hygienist are well protected from even minor, transient exposure to mercury vapour.

Research Supporting the Continued Use of Silver Amalgam

It is clear that there is no scientific data to indicate that silver amalgam poses any significant health risk for patients. It is equally clear that there is an impressive body of research that demonstrates that amalgam is completely safe.

Two very important randomized controlled clinical trials were published in the *Journal of the American Medical Association* in 2006.[13,14] One was called the New England Children's Amalgam Trial and the other was the Casa Pia Children's Trial. Both studies included over 500 children who received multiple restorations of either composite resin or amalgam. One of the trials was 5 years in duration, and the other was 7 years. They measured subtle neurobehavioural effects to determine if there were any negative effects due to amalgam. Both studies basically found that there were no neurobehavioural effects that could be attributed to amalgam. In fact the very scant data suggested that composite resin may have more effects than amalgam, but the data were of very low scientific validity.[15–17]

The studies also showed that composite resin restorations needed to be repaired or replaced 7 times more than amalgam restorations and that composite resin restorations were 2.5 times more likely to have recurrent caries.

Environmental Impact

Concerns have been raised regarding the impact mercury in amalgam has on the environment. Major environmental sources of mercury include volcanoes (50%), coal combustion (33%) and gold production (6%). Amalgam's only contribution in this area is in waste disposal, largely through mercury resulting from crematories, and is less than 1% of the total. Many jurisdictions require dental offices to utilize sophisticated amalgam separators, and a 2016 FDA regulation requires all offices in the United States to install such separators within 3 years.

In January 2013, a number of nations signed the Minamata Convention in Geneva. This meeting was convened to discuss the influence of mercury on health and the environment. Discussions on amalgam were a small part of that meeting, and there were no curbs imposed on the use of silver amalgam. The group declared that the use of amalgam should be reduced in coming years through improved preventive measures and that research should be encouraged to develop materials that will work as good or better than amalgam. In philosophic terms, the threat that may eliminate amalgam use as a restorative material sometime in the future is not a question of toxicity to humans but, rather, of *environmental protection*.

An analysis of the data presented in this section leads to the conclusion that *mercury in silver amalgam restorations poses absolutely no problem for dental patients*. This conclusion has been reached by experts in the field, by consumer interest groups and by thorough reviews by governmental agencies.

Clinical Notes

I. Small amounts of mercury vapour are released from the surface of silver amalgam restorations during mastication.
II. The amount of mercury vapour released is very small, and poses no risk to the integrity of the restoration or to the systemic health of the patient.
III. There are virtually no published scientific papers validating the belief that mercury in silver amalgam is potentially dangerous to patients.
IV. There is an ample and growing body of research that demonstrates the safety of silver amalgam.
V. This research plus the over 100-year history of use of the material supports its continued use.
VI. Although mercury in silver amalgam poses no health risk to dental patients, even in dental offices where excellent mercury hygiene is practiced, dental personnel are occupationally exposed to higher levels of mercury vapour than the general population. This is due to vapour that is released when amalgams are placed and, even more importantly, when they are removed, or when teeth with amalgam core build-ups are prepared for crowns. Although the levels of mercury vapour to which dental personnel are routinely exposed are well above those of the general population, they are generally below the current TLV. This occupational exposure to mercury vapour poses no risk as long as proper procedures for mercury hygiene are utilized. All dental personnel should be familiar with the procedures essential for the proper practice of mercury hygiene.

Summary

Amalgam is a safe and effective direct restorative material. An amalgam restoration is relatively easy to accomplish, and adherence to tooth preparation and material handling requirements results in clinical success. Although the scope of the clinical uses of amalgam presented in this book is narrower than in the past, amalgam still is recognized as an excellent material for restoring many defects in teeth.

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